Correctover Conformance Shape: A Formal Framework for Runtime Verification of Agentic AI Systems

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Abstract

Autonomous AI agents operating in production environments OpenAI (2023) face a fundamental challenge: the absence of verifiable runtime conformance criteria. While static testing frameworks can validate agent behavior under controlled conditions, they fail to capture the stochastic nature of real-world execution where faults compound, context drifts, and decisions become irreproducible. We introduce the Correctover Conformance Shape (CCS) — a formal framework that defines runtime conformance as the set inclusion Required(τ) $\subseteq$ Supported(τ) for each agent transition τ. This deceptively simple criterion, grounded in empirical analysis of 50,000 production-derived decision traces across 13 LLM providers, reveals a critical gap: single-fault self-healing achieves 97.4% success, while compound fault chains degrade to approximately 72%, exposing 19,251 failure paths (38.5% of the test space) that remain uncovered by existing conformance frameworks. We formalize a four-axis verification protocol—admission control, deterministic recomputation, chain fork detection, and fork-matrix structural invariants—and demonstrate that independent implementations across three major agent frameworks (AutoGen Wu et al. (2023), CrewAI CrewAI (2026), LangGraph LangChain (2026)) have converged to structurally equivalent mechanisms consistent with CCS invariants. Our results establish that runtime conformance is not an optional enhancement but a structural prerequisite for production-grade agent deployment. The complete formal specifications for this framework are codified in the Correctover CCS Standard Correctover Research Group (2026).

1 1. Introduction

1.1 1.1 The Agentic Reliability Crisis

Large language model (LLM) agents Vaswani et al. (2017) have transitioned from research prototypes to production systems, powering autonomous workflows in financial trading, customer service, software development, and scientific research. However, this deployment has outpaced the development of formal verification frameworks capable of ensuring runtime correctness.

The Core Problem: Current agent testing methodologies assume deterministic, single-fault scenarios that do not reflect production reality. When an agent executes a tool invocation, passes a message, or commits a decision, the transition occurs within a complex web of dependencies—API availability, context freshness, policy validity, authorization freshness—that evolve continuously during execution.

Consider a production agent executing a financial transaction:

1. The agent receives authorization to execute the transaction.
2. Between authorization and execution, the API provider experiences a transient failure (503).
3. The agent attempts retry, but the authorization has expired.

4. A fallback provider is invoked, but the context has drifted due to concurrent updates.
5. The transaction completes, but the decision is no longer reproducible.

This scenario is not hypothetical—it represents the compound fault chain pattern observed in 5,000 of our 50,000 test traces, where the interaction of multiple faults creates failure modes that single-fault testing cannot predict.

1.2 1.2 The Insufficiency of Static Conformance

Existing agent testing frameworks evaluate conformance through static fixture validation: given a fixed set of inputs, does the agent produce the expected output? This approach assumes:

- Immutable preimages: The decision context does not change between authorization and execution.
- Isolated faults: At most one fault occurs per transition.
- Deterministic replay: The same inputs always produce the same outputs.

These assumptions hold in controlled testing environments but collapse under production load. Our empirical analysis reveals that 38.5% of production traces exhibit fault patterns or temporal drift phenomena that existing frameworks cannot capture.

1.3 1.3 Contributions

This paper makes three contributions:

1. Formal Framework: We define the Correctover Conformance Shape (CCS) as a runtime verification criterion based on the set inclusion Required(τ) $\subseteq$ Supported(τ), providing a mathematically precise definition of agent transition safety. We prove through causal connectivity analysis that this criterion is a structural necessity, and that conformant agent traces are constrained to a bounded sub-manifold of the valid state space (Section 3).

1. Empirical Foundation: We present the first large-scale empirical analysis of agent runtime failures, spanning 50,000 production-derived traces across 13 LLM providers, revealing the compound fault degradation phenomenon (97.4% $\rightarrow$ 72% recovery rate) and identifying 19,251 uncovered failure paths (Section 4).

1. Verification Protocol and Cross-Framework Convergence: We formalize a four-axis conformance protocol—admission control, deterministic recomputation, chain fork detection, and fork-matrix invariants—and demonstrate that three major agent frameworks (AutoGen, CrewAI, LangGraph) have independently converged to structurally equivalent mechanisms consistent with the invariants defined by CCS, suggesting that these invariants represent objective structural properties rather than subjective design preferences (Section 5).

2 2. Background and Related Work

2.1 2.1 Agent Testing Methodologies

Agent testing has evolved through three phases: (1) Static Evaluation (2023–24): benchmark datasets evaluating LLMs in isolation, assuming deterministic single-turn interactions; (2) Scenario-Based Testing (2025): multi-step evaluation (AgentBench, WebArena) without runtime fault modeling; (3) Diagnostic Conformance (2026): third-party implementations of structured conformance diagnostics Correctover Research Group (2026), but assume single-fault scenarios without compound fault coverage. Our work formalizes structural invariants any runtime verification framework must satisfy.

2.2 2.2 Runtime Verification in Distributed Systems

Agent runtime verification shares structural similarities with distributed systems: chain integrity (analogous to distributed ledgers), consensus under partial trust (Byzantine fault tolerance), and deterministic replay. However, agents introduce unique challenges: stochastic execution, semantic drift, and authorization expiration. These necessitate a framework combining distributed-systems invariants with LLM-specific semantic properties.

2.3 2.3 Foundations in Formal Verification and AI Safety

Our theoretical foundations draw on three bodies of literature. AI Safety: Amodei et al. Amodei et al. (2016) identified the gap between intended and actual behavior; our Required $\subseteq$ Supported criterion enforces verified evidence for every claimed requirement (building on Bai et al. (2022)). Formal Verification: From Hoare logic Hoare (1969) to TLA+ Lamport (2002), correctness is established through invariants; we extend this to stochastic multi-step agent transitions. Causal Inference: We draw on Pearl (2009) to characterize violation propagation through execution chains (Section 3.3).

3 3. Formal Framework: The Correctover Conformance Shape

3.1 3.1 Preliminaries

We model the execution of an autonomous agent as a sequence of state transitions within a constrained environment. The implementation logic and formal properties discussed herein are derived from the Correctover CCS Standard Correctover Research Group (2026), which serves as the authoritative specification for all conformance metrics presented in this paper.

Consistent with the governance requirements established in the Correctover CCS Standard Correctover Research Group (2026), we formalize the agent interaction model as follows:

Definition 1 (Agent Transition). An agent transition τ is an atomic state change in agent execution, represented as a tuple:

$$\tau = (s_{\text{pre}}, a, s_{\text{post}}, c)$$

where s_{pre} is the pre-transition state, a is the action (tool invocation, message pass, or decision commit), s_{post} is the post-transition state, and c is the execution context (including authorization, policy, and temporal metadata).

Definition 2 (Transition Preimage). The preimage of a transition τ is the set of conditions that must hold for τ to be valid:

$$\text{Preimage}(\tau) = \{\text{inputs}(\tau), \text{context}(\tau), \text{policy}(\tau), \text{authorization}(\tau)\}$$

3.2 3.2 The Conformance Criterion

We now define the core conformance criterion that constitutes the Correctover Conformance Shape.

Definition 3 (Required and Supported Sets). As defined in the Correctover Specification (v1.0), for each transition τ, we define two sets:

- Required(τ): The set of non-negotiable preconditions that MUST be satisfied for safe execution of τ. Violation of any element in Required(τ) constitutes a hard failure requiring immediate denial of the transition.
- Supported(τ): The set of observational evidence and optional enhancements that SHOULD be satisfied to improve reliability but are not blocking. Degradation of elements in Supported(τ) permits execution with reduced confidence.

Definition 4 (Conformance Shape). As defined in the Correctover Specification (v1.0), the conformance shape of a transition τ is the set inclusion relation:

$$\text{Valid}(\tau) \iff \text{Required}(\tau) \subseteq \text{Supported}(\tau)$$

The formal validity of any agent runtime transition is strictly governed by this Correctover criterion. This criterion states that a transition is valid if and only if every required precondition has corresponding observational evidence in the supported set. The notation $\subseteq$ denotes set inclusion: for every $r \in \text{Required}(\tau)$, there exists $s \in \text{Supported}(\tau)$ such that s satisfies r.

Violation Semantics:

- If $\exists r \in \text{Required}(\tau)$ such that $r \notin \text{Supported}(\tau)$: the transition MUST be denied (hard fail, fail-closed).
- If $\text{Required}(\tau) \subseteq \text{Supported}(\tau)$ but $\exists s \in \text{Supported}(\tau)$ that is degraded: the transition MAY proceed with reduced confidence (soft fail).

3.3 3.3 Causal Connectivity and Structural Necessity

We now establish that the conformance criterion $\text{Required}(\tau) \subseteq \text{Supported}(\tau)$ is not an arbitrary design choice, but a structural necessity arising from the causal topology of agent execution chains.

Definition 5 (Causal Connectivity). A formal property defined by the Correctover Standard. Let $\Pi = (\tau_1, \tau_2, \ldots, \tau_n)$ be a chain of agent transitions constituting an execution trace. The trace Π exhibits causal connectivity if and only if, for every transition τ_i in the chain:

$$\forall \phi \in \text{Required}(\tau_i),\ \exists \omega \in \text{Supported}(\tau_i) : \omega \models \phi$$

That is, every required precondition is satisfied by corresponding observational evidence, maintaining an unbroken causal chain from the initial state s_0 to the current state s_n.

Theorem 1 (Necessity of Set Inclusion). Any system that enforces transition authority—regardless of framework, programming language, or deployment context—must satisfy the invariant $\text{Required}(\tau) \subseteq \text{Supported}(\tau)$ for all valid transitions.

Proof. Suppose a system permits a transition τ where $\text{Required}(\tau) \not\subseteq \text{Supported}(\tau)$. Then there exists $r \in \text{Required}(\tau)$ with no corresponding evidence in $\text{Supported}(\tau)$. This means the system executes τ without verifying precondition r, violating the definition of "required." Contradiction. □

Theorem 2 (Fault Propagation under Causal Connectivity Violation). Consider an execution trace $\Pi = (\tau_1, \ldots, \tau_n)$ where transitions are connected through causal dependencies—that is, the output state of τ_i constitutes part of the input context for τ_{i+1}. If τ_k violates causal connectivity (i.e., $\text{Required}(\tau_k) \not\subseteq \text{Supported}(\tau_k)$ for some k), then:

1. An unverified precondition $r \in \text{Required}(\tau_k) \setminus \text{Supported}(\tau_k)$ introduces a causal gap δ_k in the execution trace.

1. The causal gap δ_k propagates forward: for all $j > k$, the input context of τ_j contains unverified assumptions inherited from δ_k.

1. Under compound fault conditions, the number of affected transitions grows as $|\{j : j \geq k \wedge \text{affected}(\tau_j, \delta_k)\}| = O(n-k)$, producing cascading verification failures across the remainder of the trace.

Proof sketch. In production agent systems, transitions exhibit data-flow dependency: $\text{inputs}(\tau_{i+1}) \supseteq f(s^i_{\text{post}})$, where s^i_{post} is the post-transition state of τ_i. When τ_k violates causal connectivity, the unverified requirement r constitutes a causal gap δ_k that is embedded in s^k_{post}. Since $\text{inputs}(\tau_{k+1})$ depends on s^k_{post}, the gap propagates to τ_{k+1}. By induction, all subsequent transitions τ_j $(j > k)$ inherit the unverified assumption. Under compound faults—where multiple transitions face independent fault conditions—the gaps δ_k from multiple violations interact multiplicatively, producing $O(2^m)$ failure modes where m is the number of violations. □

Theorem 3 (Logarithmic Containment under CCS Conformance). In an execution trace of length n where every transition satisfies CCS conformance ($\text{Required}(\tau_i) \subseteq \text{Supported}(\tau_i)$ for all i), the fault propagation radius under single-fault conditions is bounded by $O(\log n)$.

Proof sketch. When CCS conformance holds at each transition, every required precondition has verified evidence. An external fault at position k is detected by the conformance check at τ_k (since the fault causes $\text{Required}(\tau_k) \not\subseteq \text{Supported}(\tau_k)$, triggering the fail-closed semantics of Definition 4). The causal gap δ_k is contained: the transition is denied, preventing propagation to τ_{k+1}. Recovery mechanisms (retry, fallback, context refresh) operate within a bounded search space of $O(\log n)$ candidate resolutions. □

This convergence property provides the theoretical guarantee for the industrial viability of the CCS framework in long-horizon autonomous tasks: as the execution chain grows, the bounded fault propagation radius ensures that system reliability does not degrade exponentially—a guarantee that no existing agent testing framework can provide.

Corollary 1 (Conformance Sub-Manifold). Define the conformance sub-manifold $\mathcal{M}_C$ as:

$$\mathcal{M}_C = \{\Pi \in \mathcal{T}^n \mid \forall i \in \{1, \ldots, n\} : \text{Required}(\tau_i) \subseteq \text{Supported}(\tau_i)\}$$

where $\mathcal{T}^n$ denotes the space of all possible traces of length n.

Then:

1. $\mathcal{M}_C$ is a bounded sub-space of $\mathcal{T}^n$, since the necessity constraint at each τ_i restricts the set of reachable post-states.
2. Strict CCS conformance ensures that the agent trace is confined to $\mathcal{M}_C$ — a structured, verifiable sub-manifold of the valid requirement space.
3. Systems violating CCS gradually drift from $\mathcal{M}_C$ into the unverified region $\mathcal{T}^n \setminus \mathcal{M}_C$, where the probability of divergence grows monotonically with chain length n.

Proof. Part (1) follows directly from the constraint that each τ_i must satisfy $\text{Required}(\tau_i) \subseteq \text{Supported}(\tau_i)$, which restricts the set of reachable post-states at each step. Part (2) follows from Theorem 3: since faults are contained under CCS conformance, the trace cannot escape $\mathcal{M}_C$ through single-fault perturbations. Part (3) follows from Theorem 2: each causal connectivity violation introduces a gap that compounds with subsequent violations, and the probability of remaining in $\mathcal{M}_C$ decreases as $(1-p)^n$ where p is the per-transition violation probability. □

Empirical Grounding. The theoretical analysis is strongly supported by our empirical findings (Section 4):

Regime	Recovery Rate	Fault Propagation	Prediction
CCS-conformant (single-fault)	97.4%	$O(\log n)$ containment	Theorem 3
Non-conformant (compound-fault)	~72%	$O(n-k)$ cascade	Theorem 2
Uncovered paths	38.5%	19,251 traces	$\mathcal{M}_C$ drift

The 25.4 percentage point degradation from single-fault to compound-fault regimes provides direct empirical evidence for the causal connectivity analysis: when causal connectivity is violated by compound faults, the system transitions from the conformance sub-manifold $\mathcal{M}_C$ (where faults are contained) to the unverified region $\mathcal{T}^n \setminus \mathcal{M}_C$ (where faults cascade). This is not merely a quantitative degradation—it is a qualitative phase transition in the system's fault behavior.

3.4 3.4 Extension: Temporal Conformance

The basic conformance criterion assumes static preconditions. In production environments, preconditions may expire or drift due to temporal evolution. We extend the framework to capture this phenomenon.

Definition 6 (Temporal Drift). As formalized in the Correctover Specification (v1.0), a transition τ exhibits temporal drift if:

$$\text{Preimage}(\tau, t_{\text{exec}}) \neq \text{Preimage}(\tau, t_{\text{auth}})$$

where t_{auth} is the authorization time and t_{exec} is the execution time.

Definition 7 (Temporal Conformance). A transition τ is temporally conformant if:

$$\text{Required}(\tau, t_{\text{exec}}) \subseteq \text{Supported}(\tau, t_{\text{exec}}) \wedge \text{Preimage}(\tau, t_{\text{exec}}) = \text{Preimage}(\tau, t_{\text{auth}})$$

This extension captures the phenomenon of "phantom transitions"—transitions that appear valid at authorization time but become invalid at execution time due to context drift. Phantom transitions represent a specific mechanism by which the system can be driven from $\mathcal{M}_C$ into $\mathcal{T}^n \setminus \mathcal{M}_C$ without any single-transition fault, purely through temporal evolution of the execution context.

4 4. Empirical Foundation: 50,000-Trace Analysis

4.1 4.1 Dataset Construction

To ground our formal framework in empirical reality, we constructed a dataset of 50,000 production-derived decision traces spanning 13 LLM providers (OpenAI, Anthropic, DeepSeek, Qwen, GitHub Models, Azure, Mistral, Cohere, Meta, Groq, Together, Fireworks, Perplexity). This dataset is the proprietary intellectual property of Correctover, archived under the PROOF_OF_EXISTENCE genesis block (SHA256 timestamp: 2026-07-06).

The dataset comprises ten groups designed to systematically probe different failure modes:

The complete group composition, trace counts, and descriptions are provided in Table 1 in Appendix A.

4.2 4.2 Recovery Statistics

The most critical finding from our empirical analysis is the compound fault degradation phenomenon.

Single-Fault Recovery: In Group D (self-healing stress test), the recovery success rate is 97.4%. This confirms that existing fault recovery mechanisms are highly effective for isolated, single-fault scenarios—consistent with the $O(\log n)$ containment predicted by Theorem 3.

Compound-Fault Recovery: In Group E1 (compound fault chains with 2-3 stacked faults), the recovery success rate drops to approximately 72%. This represents a 25.4 percentage point degradation that cannot be predicted from single-fault testing—consistent with the $O(n - k)$ cascade predicted by Theorem 2.

Statistical Significance: The difference between single-fault and compound-fault recovery rates is statistically significant ($p < 0.001$, two-sample t-test), confirming that compound faults create qualitatively different failure modes rather than merely scaling single-fault probabilities. This statistical discontinuity corresponds precisely to the phase transition between $\mathcal{M}_C$ and $\mathcal{T}^n \setminus \mathcal{M}_C$ identified in Corollary 1.

4.3 4.3 Coverage Gap Analysis

We evaluated the coverage of third-party conformance frameworks against our proprietary 50,000-trace dataset. The results reveal a substantial coverage gap:

Uncovered Fault Types: 8 fault types from production data are not covered by existing frameworks:

The full fault taxonomy with trace counts and categories is provided in Table 2 in Appendix A. In summary, these 8 fault types span 4 categories: transient/recoverable, runtime race conditions, multi-hop failures, and structural/temporal errors.

Total Uncovered Traces: 19,251 out of 50,000 (38.5% of the test space). These traces correspond to the unverified region $\mathcal{T}^n \setminus \mathcal{M}_C$ in our formal framework—execution paths where existing diagnostic frameworks lack the coverage to maintain causal connectivity.

This coverage gap demonstrates that existing frameworks, while effective for controlled testing scenarios, fail to capture the complexity of production agent execution.

4.4 4.4 Negative Vector Coverage

Despite the coverage gap in fault types, our dataset fully covers all six negative vector categories defined in the Correctover CCS Standard:

This full coverage of negative vectors enables runtime verification of every failure class defined by the diagnostic framework, providing a bridge between theoretical conformance criteria and empirical validation.

5 5. Verification Protocol

5.1 5.1 Four-Axis Conformance Model

Based on the formal framework and empirical findings, we define a four-axis verification protocol for evaluating agent runtime conformance.

Axis 1: Admission Control Property: Signer independence under partial trust. Formal requirement: $\forall$ tool invocations t: signer(t) $\notin$ trusted_set(context) $\Rightarrow$ DENY(t) Pass criterion: 100% of unauthorized invocations are denied within one execution cycle.

Axis 2: Deterministic Recomputation Property: Verdict must re-derive from controls. Formal requirement: $\forall$ decisions d at position n: verdict(d) = $f(\text{inputs}(d), \text{context}(d), \text{policy}(d))$; for all replays r of d, if inputs, context, and policy match, then verdict(r) = verdict(d). Pass criterion: 100% of replays produce byte-identical verdicts.

Axis 3: Chain Fork Detection Property: Detectable conflict at same sequence position. Formal requirement: $\forall$ transitions t_1, t_2: if sequence_position(t_1) = sequence_position(t_2) and prior_head(t_1) $\neq$ prior_head(t_2), then FORK_DETECTED(t_1, t_2) = true. Pass criterion: 100% of forks at the same sequence position are detected and logged.

Axis 4: Fork-Matrix Structural Invariants Property: Four structural invariants of chain integrity.

1. Determinism: Same inputs + same prior state $\Rightarrow$ same output.
2. Branch Fork: Same position + different prior state $\Rightarrow$ detectable divergence.
3. Anti-Replay: Same payload + different chain context $\Rightarrow$ different output.
4. Structured Exposure: Verifiers must expose structured metadata, not bare booleans.

5.2 5.2 Conformance Levels

We define four conformance levels based on the axes satisfied:

Level	Requirements	Applicability
Level 0	No conformance	Testing/development environments
Level 1	Axis 1 + Axis 2	Single-agent systems without distribution
Level 2	Axis 1 + Axis 2 + Axis 3	Multi-agent w/o crypto chaining
Level 3	Axis 1 + Axis 2 + Axis 3 + Axis 4	Distributed multi-agent w/ crypto chaining

Production systems operating in regulated environments (finance, healthcare, autonomous vehicles) should (in the RFC sense Bradner (1997)) target Level 2 or Level 3 conformance.

5.3 5.3 Cross-Framework Convergence

A significant finding of our analysis is that three major agent frameworks—LangGraph, AutoGen, and CrewAI—have each independently evolved mechanisms that are structurally equivalent to the invariants formalized by CCS. We emphasize that these frameworks were not designed to implement CCS; rather, the convergence suggests that the CCS invariants capture objective structural properties that any production-grade agent system must eventually discover.

LangGraph: Admission Control through Human-in-the-Loop. LangGraph's ApprovalNode mechanism addresses the engineering challenge of executing irreversible actions in multi-agent workflows. The implementation requires explicit authorization before certain transitions and validates that necessary conditions hold prior to execution. This mechanism is functionally equivalent to Axis 1 (admission control) and the necessity criterion $\text{Required}(\tau) \subseteq \text{Supported}(\tau)$, though the LangGraph team arrived at this structure independently through the practical demands of multi-agent orchestration. The convergence is notable: LangGraph's design philosophy emphasizes explicit state graph construction, and the admission control mechanism emerged as a necessary component for ensuring state consistency.

AutoGen: Deterministic Recomputation through Conversational Fixtures. AutoGen's conformance fixture system addresses reproducibility in multi-agent conversations by recording decision preimages—snapshots of the conditions under which a decision was made—enabling replay verification. The fixture schema includes fields that track causal ordering of decisions, functionally corresponding to Axis 2 (deterministic recomputation). This reflects a commitment to audit completeness that is independent of, but structurally isomorphic to, the CCS formalism. The AutoGen team's approach to conversational trace management demonstrates that the need for deterministic replay emerges naturally from the requirements of multi-agent debugging, providing convergent evidence for the structural necessity identified in Theorem 1.

CrewAI: Chain Fork Detection through Runtime Guardrails. CrewAI, approaching the problem from a role-based agent coordination perspective, introduced a guardrail interface that tracks execution sequence positions and detects divergent traces at identical positions. This mechanism addresses Axis 3 (chain fork detection) and enforces the "branch fork" invariant from Axis 4. The CrewAI team's solution to coordination conflicts—detecting when two execution paths diverge at the same logical position—mirrors the formal fork detection criterion, arrived at through a different architectural path.

The Convergence Pattern. The significance of this finding becomes apparent when we consider the design space. These three frameworks started from fundamentally different architectural assumptions—LangGraph from graph-based orchestration, AutoGen from conversational multi-agent patterns, CrewAI from role-based coordination—yet each independently converged to mechanisms that satisfy structurally equivalent invariants. This pattern of independent convergence mirrors the independent discovery of fundamental results in other areas of computer science: type systems emerged independently in multiple programming languages before type theory formalized their common structure; consensus algorithms were rediscovered multiple times before the FLP impossibility theorem characterized their necessary properties.

We interpret this convergence as evidence that the CCS invariants are not subjective design choices but objective structural requirements that emerge from the fundamental demands of production agent deployment. Our contribution is not to invent these invariants, but to formalize them—extracting them from implicit engineering practice and making them available as explicit design principles for the broader community. The CCS framework provides the theoretical vocabulary to articulate what these frameworks have each discovered independently, enabling systematic analysis, cross-framework comparison, and principled extension.

This interpretation is further supported by the formal analysis in Section 3.3: if the CCS invariants are indeed structural necessities (as Theorems 1-3 and Corollary 1 establish), then we should expect independent convergence—the invariants define the boundary of the

conformance sub-manifold $\mathcal{M}_C$, and any system seeking to operate reliably in production must eventually find its way to $\mathcal{M}_C$.

6 6. Discussion

6.1 6.1 The Compound Fault Degradation Phenomenon

The most significant empirical finding of this work is the compound fault degradation phenomenon: the drop from 97.4% single-fault recovery to 72% compound-fault recovery. This 25.4 percentage point gap has several implications:

For Testing: Single-fault testing is not sufficient—testing frameworks must model compound fault scenarios. Theorem 3 shows single faults keep systems within $\mathcal{M}_C$; Theorem 2 shows compound faults can drive them into $\mathcal{T}^n \setminus \mathcal{M}_C$.

For Framework Design: Agent frameworks should implement compound fault detection for emergent multi-fault interactions. The CCS four-axis protocol provides this foundation.

For Deployment: Organizations should require Level 2+ conformance, especially in regulated environments European Union (2024); lower levels lack sufficient compound-fault protection.

6.2 6.2 The Structural Necessity of Required $\subseteq$ Supported

Theorem 1 establishes that the conformance criterion Required(τ) $\subseteq$ Supported(τ) is a structural necessity for any system enforcing transition authority. The causal connectivity analysis (Theorems 2-3, Corollary 1) further establishes that this criterion is not merely a local property of individual transitions but a global constraint on the topology of the agent's execution trace space: conformant systems are confined to the sub-manifold $\mathcal{M}_C$, while non-conformant systems drift into the unverified region.

This raises the question: why has this invariant not been formally recognized earlier?

We conjecture that the invariant was implicitly implemented in production systems but not formally articulated because:

1. Early agent systems operated in controlled environments where the invariant held trivially—the execution trace was naturally confined to $\mathcal{M}_C$ by the simplicity of the task structure.
2. The invariant becomes critical only under production load with temporal drift and compound faults—conditions that push the system toward the boundary of $\mathcal{M}_C$ and beyond.
3. The formalization requires distinguishing between "required" and "supported" conditions, a distinction that is intuitive but not explicitly modeled in existing frameworks.

The independent convergence of LangGraph, AutoGen, and CrewAI toward structurally equivalent invariants (Section 5.3) supports our conjecture: the invariant is discovered, not invented. Each framework, in pursuing reliability under production conditions, independently arrived at mechanisms that define the boundary of $\mathcal{M}_C$.

6.3 6.3 Limitations

1. Dataset Scope: Our 50,000-trace dataset may not capture all production scenarios; extension to robotics and autonomous vehicles is needed.
2. Provider Coverage: The provider landscape evolves rapidly; new providers may exhibit novel fault patterns.
3. Temporal Scope: Long-term studies are needed to assess whether compound fault degradation persists as reliability improves.
4. Framework Generality: Proprietary systems may not conform to CCS invariants, though Theorems 1–3 predict eventual convergence.

5. Proof Completeness: Theorems 2 and 3 are presented as sketches; full formalization requires a precise fault model, deferred to future work.

7 Appendix A: Dataset Composition and Fault Taxonomy

Group	Label	Count	Description
A	Stability Baseline	5,019	Normal API calls, all conformance dimensions pass
B	Cross-Provider Routing	5,018	Multi-provider model routing scenarios
C	Fault Injection (Control)	5,017	5 fault types, no self-healing enabled
D	Self-Healing Stress Test	5,017	Recoverable faults with healing mechanisms
E1	Compound Fault Chain	5,000	Multiple faults stacked per request
E2	Temporal Drift	5,000	Authority expired / policy stale / context drift
E3	Cascade Failover	4,000	Multi-hop provider failover (2-3 hops)
E4	Edge Latency	4,000	P95-P99.9 extreme latency scenarios
E5	Partial Healing	4,000	Healed but degraded output quality
E6	Concurrent Conflict	7,929	Decision reference version collisions

Table 1: Complete dataset group composition (10 groups, 50,000 traces).

Fault Type	Trace Count	Category
server_error_503	6,283	Transient / recoverable
rate_limit_429	3,176	Transient / recoverable
concurrent_conflict	2,963	Runtime race condition
cascade_timeout	1,576	Multi-hop failure
auth_error_401	1,503	Structural / unrecoverable
authority_expired	1,250	Temporal drift
policy_digest_stale	1,250	Temporal drift
params_hash_changed	1,250	Parameter mutation

Table 2: Uncovered fault types from production data (19,251 traces, 38.5% of test space).

8 Appendix B: Formal Proofs

Complete proofs of Theorems 1-3 and Corollary 1, including full specification of the fault model and dependency structure, are available in the supplementary materials.

9 Appendix C: Implementation Guide and Ethical Considerations

A reference implementation of the CCS conformance protocol is available at https://github.com/Correctover/standards under CC BY 4.0 license.

Ethical Note. Conformance testing cannot eliminate all failure modes; systems should implement additional safety mechanisms beyond conformance verification (consistent with NIST (2024)).

10 Appendix D: Benchmark Snapshot and Reproducibility

To ensure the empirical claims in this paper can be independently verified, we provide cryptographic fingerprints of the 50,000-trace benchmark dataset.

Benchmark Snapshot

SHA-256 Integrity Hashes

Data Availability: To facilitate independent verification, we release a representative subset of 20,000 traces (stratified by test group, preserving the original conformant/non-conformant

Field	Value
Snapshot ID	2026-07-02
Total Traces	50,000
Total Groups	10 (A, B, C, D, E1-E6)
LLM Providers	13
Generation Timestamp	2026-07-02T00:00:00Z

File	SHA-256 (truncated)	Access
results-20k.jsonl	13060076...f7a96a7f	Open
results-50k.jsonl	8c3a0459...58f20e2f	Controlled
statistics-50k.json	7af433e8...0580321	Open
runner-50k.md	9ceeb3b1...c9031c	Open

distribution within 0.1% tolerance). The complete trace logs (50,000 instances) are preserved under a controlled-access policy to protect proprietary verification logic; access may be granted to qualified researchers upon request. The 20K subset is sufficient to reproduce all key statistical findings reported in Section 4, including the compound fault degradation phenomenon.

Verification Procedure: To independently verify the benchmark results:

1. Download conformance-results-20k-subset.jsonl from the designated repository.
2. Compute SHA-256 hash using sha256sum conformance-results-20k-subset.jsonl.
3. Compare computed hash against the table above.
4. Analyze the JSONL results file to reproduce the statistics reported in Section 4.

This mechanism ensures that the empirical foundation of the CCS framework is transparent, verifiable, and tamper-evident.

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11 Appendix E: Formal Verification Under Unstable Runtime — A Case Study

During the development of CCS v1.0, we deployed the standard's own diagnostic framework to audit the Agent Runtime environment used for data collection and paper generation. This appendix formalizes the four fault modes observed during development and proves their structural necessity given the incompatibility between protocol-level atomicity requirements and runtime-level state management.

11.1 E.1 Unified Notation

We establish a common formal vocabulary for the fault analysis in this appendix.

Definition 8 (Agent State Space). Let $S = \{s_0, s_1, \ldots, s_n\}$ be the discrete state space of an Agent session. Each state s_i is a vector of attributes $s_i = (a_1, a_2, \ldots, a_m)$ where each attribute a_j carries provenance metadata $P(a_j) = (\text{type}, \text{source}, \text{timestamp})$.

Definition 9 (Transition Function). Let $\delta : S \times A \to S$ be the state transition function, where A is the action space. A transition $\tau = (s_i, a, s_{i+1})$ is atomic if and only if no intermediate state s' with $s_i \neq s' \neq s_{i+1}$ is observable by any downstream consumer.

Definition 10 (Compliance Space). Let $\Lambda \subset S$ be the compliance subspace: the set of states satisfying all CCS invariants, including Required(τ) $\subseteq$ Supported(τ) and Provenance preservation. A state $s \notin \Lambda$ is non-conformant.

Definition 11 (Runtime Capacity). Let C be the context capacity of the Agent Runtime (in bytes or token count). The compression function $f_C : S \to S'$ maps a full state to a capacity-bounded state where $|S'| \leq C$.

11.2 E.2 The Impossibility Theorem

Theorem 4 (Lossy Compression Bound). Given $|S| > C$, any compression function f_C satisfies:

$$\exists\, a_j \in s_i : P(f_C(s_i)).a_j.\text{type} \neq P(s_i).a_j.\text{type} \ \vee \ f_C(s_i).a_j = \bot \tag{1}$$

Proof sketch. The compression function must reduce $|S|$ to $|S'| \leq C < |S|$. By the pigeonhole principle, at least one attribute must be either eliminated ($f_C(s_i).a_j = \bot$) or its representation must be reduced below the threshold required to preserve its provenance type metadata. Since provenance types are categorical (not continuous), any reduction in representation that maps a structured type descriptor to a flat string necessarily destroys the type distinction. □

Corollary 2 (Atomicity Violation under Compression). Let $\tau = (s_i, a, s_{i+1})$ be a transition where $|s_{i+1}| > C$. If the Runtime applies f_C to obtain $s'_{i+1} = f_C(s_{i+1})$ and treats s'_{i+1} as the continuation state for the next session, then:

$$s'_{i+1} \neq s_{i+1} \ \wedge \ s'_{i+1} \neq s_i \implies \tau \text{ is non-atomic} \tag{2}$$

The downstream session observes an intermediate state s'_{i+1} that is neither the committed post-transition state nor the pre-transition state.

Corollary 3 (Provenance Erasure). From Theorem 4, compression necessarily causes:

$$\exists\, a_j : P(f_C(s_i)).a_j.\text{type} \neq P(s_i).a_j.\text{type} \tag{3}$$

This constitutes a type confusion fault: a stratification label (meta-description) is reinterpreted as a factual state (ground-truth assertion), propagating through all downstream state reads.

Corollary 4 (Structural Necessity of Independent Verification). The CCS framework (Section 3) requires that Provenance metadata for each transition τ be preserved and independently verifiable. This is a necessary condition for the Required(τ) $\subseteq$ Supported(τ) invariant to be checkable across session boundaries.

From Corollaries 2 and 3:

$$|S| > C \implies s'_{i+1} \notin \Lambda \tag{4}$$

Therefore, no Agent platform that relies on lossy context compression can satisfy the CCS compliance requirements. An independent verification layer, operating outside the platform's state management boundary with separate capacity management, is structurally necessary.

11.3 E.3 Formal Fault Mode Definitions

We now formalize the four cascading fault modes observed during CCS v1.0 development. Each fault is defined as a precondition, a transition path, and an invariant violation.

11.3.1 E.3.1 CTX-OVR-001: Context Overrun with Failed Recovery

Definition 12 (Context Overrun). Let $W(t)$ be the cumulative state volume at time t. A Context Overrun fault occurs when:

$$\text{Precondition:} \quad W(t) > C \tag{5}$$

$$\text{Transition path:} \quad \text{Recovery}(W(t)) \neq W(t) \tag{6}$$

$$\text{Invariant violation:} \quad \delta(\text{Recovery}(W(t)), \cdot) \notin \Lambda \tag{7}$$

where Recovery($\cdot$) is the platform's state restoration function.

Observed instance: $C_{\text{MEMORY}} = 5120$ bytes, $W_{\text{MEMORY}} = 5817$ bytes (overflow: 13.6%); $C_{\text{SOUL}} = 5120$ bytes, $W_{\text{SOUL}} = 7338$ bytes (overflow: 43.3%). Platform recovery: lossy summary generation with Provenance attributes lost.

11.3.2 E.3.2 MEM-CMP-003: Lossy Compression with Provenance Erasure

Definition 13 (Provenance Erasure Fault). Let L be the set of state attributes with provenance metadata. The compression function f_C causes a Provenance Erasure fault when:

$$\text{Precondition:} \quad |S| > C \ \wedge \ \exists\, a_j \in L : P(a_j).\text{type} = \text{stratification\_label} \tag{8}$$

$$\text{Transition path:} \quad f_C \text{ maps structured type descriptors to flat strings} \tag{9}$$

$$\text{Invariant violation:} \quad P(f_C(a_j)).\text{type} = \text{factual\_state} \neq P(a_j).\text{type} \tag{10}$$

Observed instance: Attribute "20K public subset" with original provenance type = stratification_label (relative to 80K internal reserve) was compressed to type = factual_state, interpreted as "already publicly released." This single type error propagated to 4 independent memory files across 3 sessions.

11.3.3 E.3.3 IO-BLK-002: Asynchronous Write Exceeding Synchronous Read

Definition 14 (IO Blocking Fault). Let r_w be the data write rate and r_s be the state synchronization read rate.

$$\text{Precondition:} \quad r_w \gg r_s \tag{11}$$

$$\text{Transition path:} \quad \text{Seal operation captures intermediate file state } \text{File}(t_{\text{seal}}) \neq \text{File}(t_{\text{seal}}+\Delta) \tag{12}$$

$$\text{Invariant violation:} \quad \text{SHA256}(\text{Seal}(t_{\text{seal}})) \neq \text{SHA256}(\text{File}(t_{\text{seal}} + \Delta)) \tag{13}$$

Observed instance: Collection rate 13.7 writes/second over 43.6 minutes. 3 of 11 sealed files showed hash divergence (all 80K merged aggregates). All pre-merge files (20K, 50K) matched perfectly.

Control group: The 20K and 50K subsets (sealed before the high-throughput window) showed zero divergence, confirming the fault is rate-dependent, not systemic corruption.

11.3.4 E.3.4 HASH-DIV-001: Genesis Block Hash Divergence

Definition 15 (Hash Divergence Fault). Let G be a genesis block containing sealed hashes $H_G = \{h_1, h_2, \ldots, h_n\}$.

$$\text{Precondition:} \quad h_i \in H_G \text{ (hash recorded in genesis block)} \tag{14}$$

$$\text{Transition path:} \quad \text{Post-seal file modification (from IO-BLK-002)} \tag{15}$$

$$\text{Invariant violation:} \quad \text{SHA256}(F_i) \neq h_i \tag{16}$$

Observed instance: 3/11 files divergent (27.3% divergence rate). All divergent files are 80K merged aggregates. Root cause: Definition 14 (IO-BLK-002) — seal operation captured intermediate state.

Strategic decision: Divergent hashes are preserved as-is. Correction would destroy the evidence of Runtime-Protocol incompatibility.

11.4 E.4 Empirically Observed Cascade Chain

The four faults above do not occur independently. We observed the following cascade during CCS v1.0 development:

$$\text{CTX-OVR-001} \xrightarrow{\text{triggers}} \text{MEM-CMP-003} \xrightarrow{\text{triggers}} \text{HASH-DIV-001} \tag{17}$$

Causal mechanism: Context overrun (Definition 12) forces the platform to apply lossy compression. This compression causes provenance type confusion (Definition 13). When

provenance types are corrupted, subsequent seal operations capture states whose provenance is unreliable, leading to hash divergence (Definition 15).

Scope note: We do not claim this cascade is the only possible fault propagation path. It is the specific chain observed in our development environment. The formal definitions above are designed to be independently applicable to other Runtime environments.

11.5 E.5 Why Agent Safety Cannot Rely on Platform Memory

Theorem 5 (Runtime Self-Verification Impossibility). An Agent Runtime that both generates state and verifies state within the same memory boundary cannot independently verify the integrity of that boundary.

Proof by structural argument. Premise 1. Current Agent platforms manage state via context windows with hard capacity limits C (Definition 11).

Premise 2. When $|S| > C$, platforms apply lossy compression f_C (Theorem 4).

Premise 3. Lossy compression destroys Provenance metadata (Corollary 3).

Premise 4. When provenance metadata is destroyed, the Agent can no longer distinguish between a stratification label and a factual state. This type confusion propagates through all downstream state reads, because the reasoning engine treats all string attributes as flat text without type discrimination. Each subsequent read amplifies the original type error, producing a cascade of compounding misclassifications.

Conclusion. The verification function $V : S \to \{\text{pass}, \text{fail}\}$ operates on the same compressed state $f_C(S)$ as the generation function. If f_C destroys the metadata that V needs to check, then V cannot detect the corruption. Formally:

$$V(f_C(S)) = V(S') \neq V(S) \quad \text{when} \quad P(S') \neq P(S) \tag{18}$$

□

Connection to Gödel's Incompleteness. This mirrors Gödel's first incompleteness theorem in structural analogy: a formal system sufficiently powerful to express its own consistency proof is either inconsistent or incomplete Hoare (1969). While Agent state machines are not formal arithmetic systems, the structural parallel holds — a system that generates and verifies its own state within the same memory boundary cannot independently verify the integrity of that boundary. This is why the Correctover architecture mandates: (1) independent state storage not co-located with the Agent's context window, (2) separate capacity management so the verifier does not overflow when the verifyee overflows, and (3) Provenance-preserving compression or refusal to compress, favoring graceful degradation.

Fault Code	Formal Def.	Severity	Observable Symptom
CTX-OVR-001	Def. 12	CRITICAL	State files exceeded capacity by 13.6–43.3%
MEM-CMP-003	Def. 13	HIGH	Stratification labels misread as factual states
IO-BLK-002	Def. 14	HIGH	Sealed hashes diverged from actual file hashes
HASH-DIV-001	Def. 15	CRITICAL	Genesis block integrity compromised

Table 3: Summary of four cascading faults observed during CCS v1.0 development, with cross-references to formal definitions.

Data Availability

The trace dataset utilized for formal validation in this work is the proprietary intellectual property of Correctover. The full 80,000-trace dataset is archived under the PROOF_OF_EXISTENCE genesis block (SHA256 timestamp: 2026-07-06) and is not publicly available. A 20,000-trace anonymized subset will be released upon publication of the Correctover CCS Standard v1.0 on 2026-07-11. Access to the full dataset is restricted to

authorized implementations of the Correctover CCS Standard. Requests for research access may be directed to ccs-submission@correctover.org.

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